

LOGITECH DEVSTUDIO CHALLENGE 2025 • STYLUS IN XR CATEGORY

NeuroSketch XR 2.0

*Not just a training tool. A new way
humanity understands the brain.*

The World's First AI-Powered
Mixed Reality Neurosurgical
Intelligence Platform.



THE SILENT CRISIS INSIDE MEDICINE

250,000

Americans die annually

from preventable medical errors.
Source: Institute of Medicine



7+ Years

Residency duration

before a neurosurgeon touches
a real brain independently.



98%

Of medical XR apps

use game controllers—not
precision instruments.



*"We send pilots into simulators before they fly. We send surgeons
into real brains with a textbook and a prayer."*

WHY NOTHING HAS FIXED THIS YET

THE OLD WAY

- ❌ 2D textbooks and flat diagrams 
- ❌ Cadavers (Static, expensive, cannot bleed)
- ❌ VR apps with Game Controllers
Like learning surgery with a TV remote 
- ❌ Geography-locked training 



THE SOLUTION

- ✅ 3D spatial hands-on interaction 
- ✅ Living physiology that responds to touch 
- ✅ Pressure-sensitive instrument feel 
- ✅ Global access—zero geography barrier 

The missing piece was never the software. It was the right hardware.

LEARN



Explore the living brain at any scale—hemisphere to synapse.

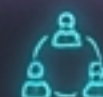
NeuroSketch XR 2.0

SIMULATE



Practice real procedures with AI that thinks, bleeds, and surprises you.

COLLABORATE



Train with experts anywhere on Earth in shared MR space.

Where the MX Ink stylus becomes the world's most sophisticated virtual surgical instrument.

FIVE PILLARS. ONE PLATFORM. ZERO COMPETITORS.



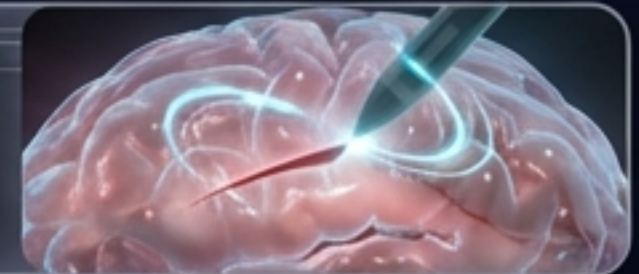
NEURAL EXPLORER

Zoom from hemisphere to synapse. Bioluminescent neural firing. Personal MRI import.



SURGICAL SIMULATOR

MX Ink = Scalpel. 12 full procedures. Pressure-mapped tissue across 14 types.



GLOBAL OPERATING ROOM

6 users. Any city. One shared holographic brain. Role-based simulation.



PATHOPHYSIOLOGY AI

Paint disease. Watch it grow in real time. Treat it and see consequences.



MASTERY SYSTEM

Surgical passport. Global leaderboards. Adaptive AI difficulty.



EVERY COMPETITOR BUILT FOR CONTROLLERS. WE BUILT FOR THIS.

4,096 Pressure Levels

Light (surface) → Medium (dissection) → Heavy (bone cut).

Tilt Detection (0-60°)

Flat = spatula retractor;
Upright = bipolar forceps.

Precision Tip

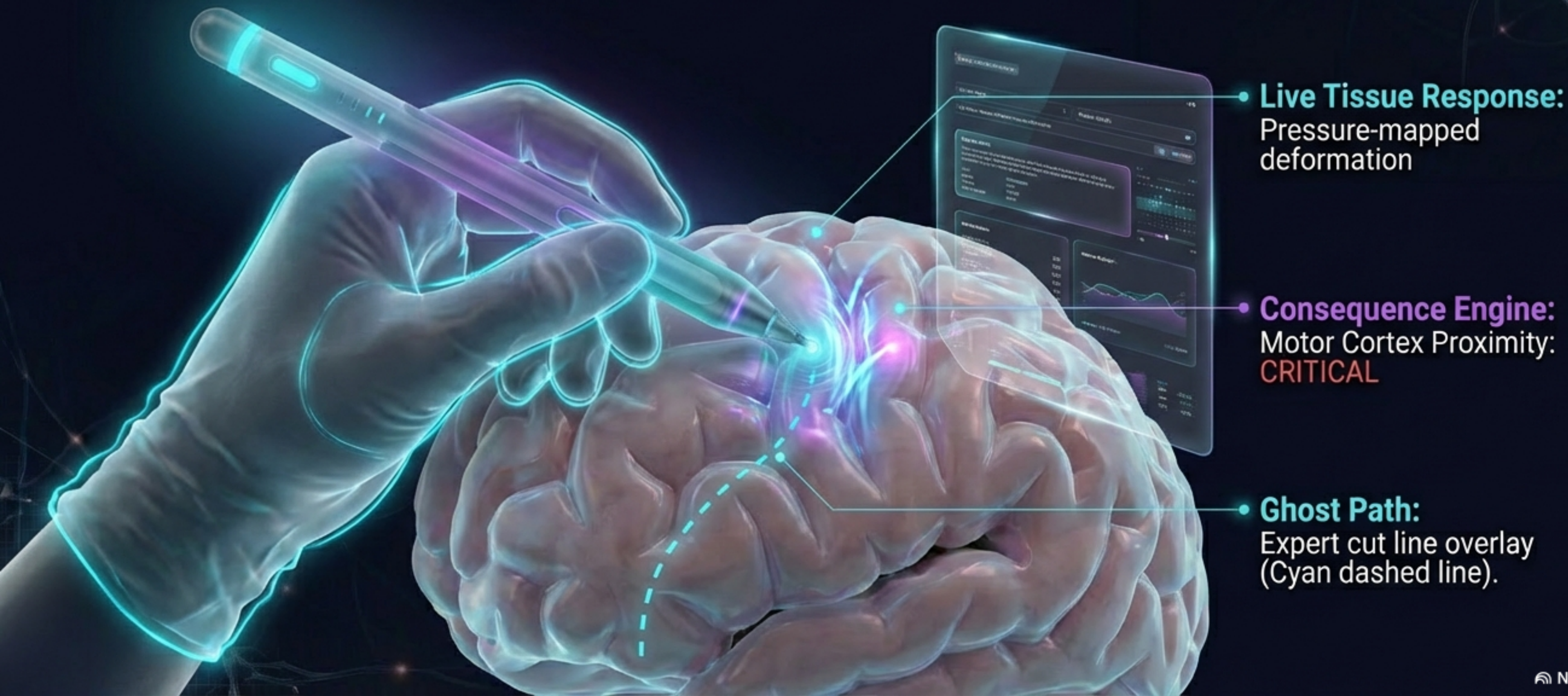
Sub-millimeter accuracy. Ghost surgeon deviation scored in real millimeters.

Side Button

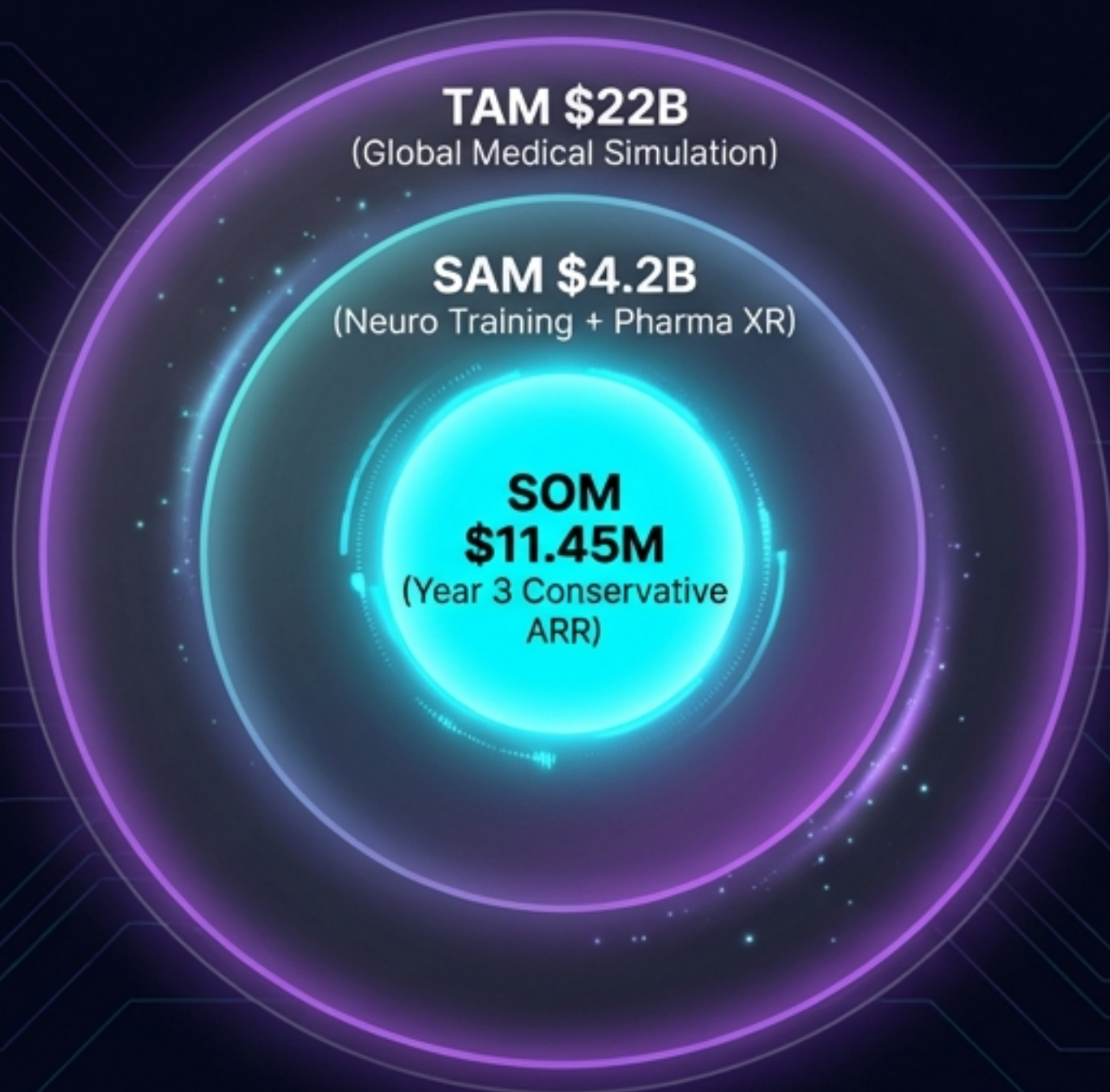
Cycle modes (Annotate → Dissect → Paint Pathology).

Feature	Game Controller	MX Ink
Pressure Sensitivity	None	4,096 Levels
Surgical Feel	Button Press	Natural Pen Grip
Skill Transfer	Zero	Direct Motor Memory

THIS IS WHAT LEARNING NEUROSURGERY FEELS LIKE IN 2025.



A \$22 BILLION MARKET WAITING FOR THIS MOMENT

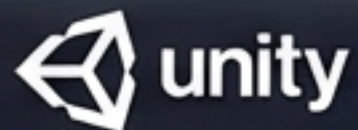
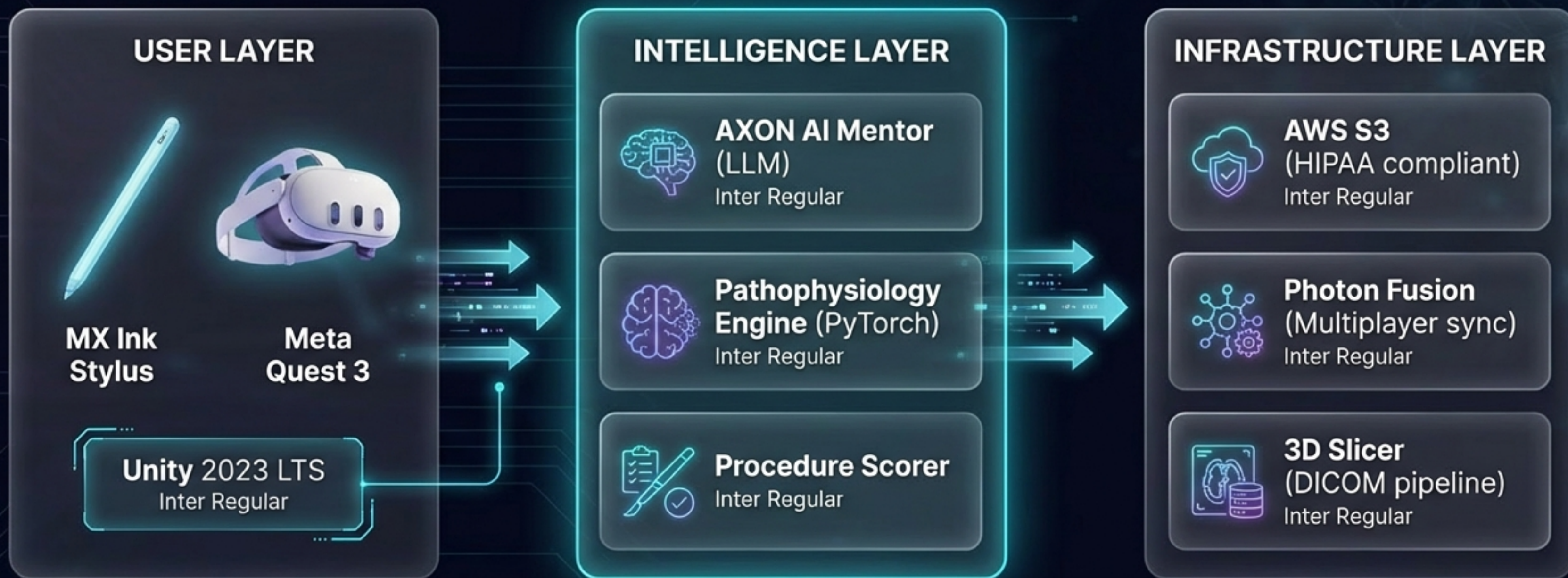


"Every zip code that has a **medical school** is our customer."

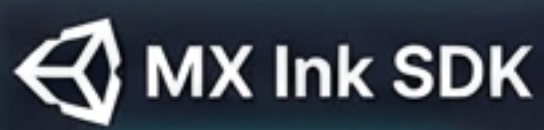
BUILT TO SCALE. FIVE REVENUE STREAMS.



PRODUCTION-GRADE ARCHITECTURE.



Meta XR SDK



PyTorch



JUDGED ON FOUR CRITERIA. DOMINANT ON ALL FOUR.

NOVELTY (10/10)

★★★★★



Zero competitors combine pressure-sensitive stylus input with AI pathophysiology.

IMPACT (10/10)

★★★★★



Addresses 250,000 preventable deaths. Democratizes training globally.

VIABILITY (9/10)

★★★★★



Clear B2B SaaS model. High-margin Pharma expansion. \$11M+ ARR potential.

UX QUALITY (10/10)

★★★★★



Every interaction is MX Ink native. The stylus IS the product.

Every other medical XR app gave surgeons a game controller. We gave them a scalpel.

EXECUTION ROADMAP: FROM MVP TO SCIENCE FICTION

THE 6-WEEK HACKATHON SPRINT




**Wk 1-2: MX Ink
Integration**
Inter Regular


**Wk 3-4: Tissue
Engine & AXON AI**
Inter Regular


**Wk 5-6: Polish
& Demo Loop**
Inter Regular

THE XR+ FUTURE

Haptics



Haptics:
Vibration feedback
on nerve proximity
Inter Regular

Time-Lapse

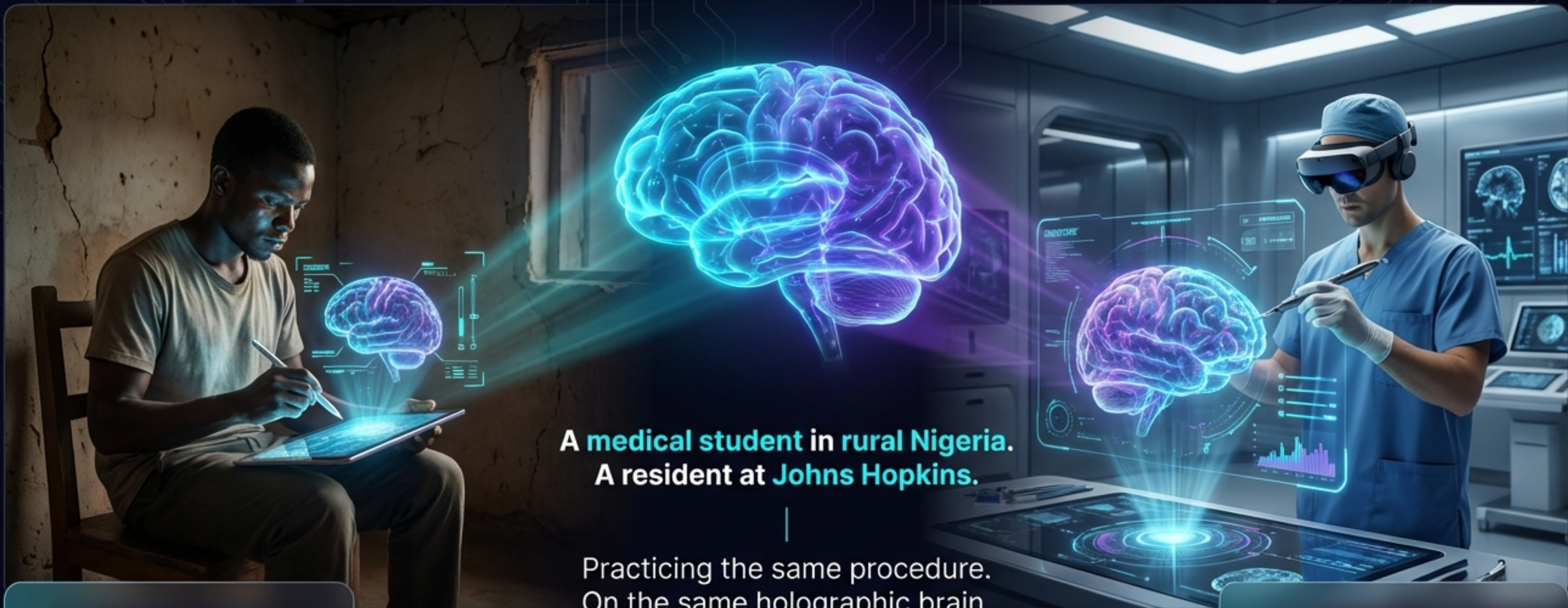


Time-Lapse:
Rewind/Fast-forward
tumor growth
Inter Regular

Pharmacology



Pharmacology:
Inject virtual drugs,
watch vessels react
Inter Regular



A medical student in rural Nigeria.
A resident at Johns Hopkins.

Practicing the same procedure.
On the same holographic brain.

For the first time, **zip code does not**
determine **surgical competence.**

THE ASK

-  Top 50 Selection
-  Logitech Hardware
-  Expert Mentorship

THE VALUE

The defining medical
use case for MX Ink.

Train like a surgeon. Think like a scientist. Save lives.